

Some notes for future team members, in case I never get around to correcting/updating this report for internal rocketry use. These will make more sense after reading the report.

- There are probably some people I forgot to acknowledge in the acknowledgements. The surface finish research project is one of my favorite things I've worked on for the team and it was thanks to a lot of people, not all of whom were listed, that it got done on time and worked as well as it did. Thanks to everyone who helped.
- Some of the stuff in there is kind of random bullshit to flesh out the work term report, e.g. don't read too seriously into the fluid dynamic explanations of things because they're probably not very accurate or relevant.
- The research happened in a much less linear fashion than the report implies.
- The OpenRocket sims described in the report were all retroactive and done by me on the day the report was due. We actually used whatever was the most recently available LOTS design cycle data (probably cycle 2) to see the effects of the surface finish, not the KOTS OpenRocket file as described in the report. This is because I was super rushed writing the report and could not find the cycle 2 sim files.
- I do genuinely recommend that the team continues to use smooth paint and not epoxy or vinyl if it is desired to reduce the rocket drag a lot. This costs basically nothing to implement and is a way more robust and practical method than the other two IMO.
 - Of course you don't need to do any of this if the engine performance or dry mass is good enough. We ran out of time to characterize samples besides 2000 grit, but it might be useful to do this research in the future so that we can accurately predict apogee on rockets where we don't want to do as much sanding.
- Some of the things described in the sample prep methodology are kind of made up because we forgot to record them (e.g. paint dilution ratios, intermediate coat sanding grit values, number of coats, etc.). But this shouldn't affect the end result much; so long as it is relatively even and sanded to 2000 grit, the surface will likely be around 1 μm .
- Unlike what the conclusion says, it might actually be sufficient to just pick a different value of roughness height to use instead of 5 μm in order to get 'close enough' to the actual flight apogee of LOTS. I was planning to do this in OpenRocket myself but ran out of time so I made up some bullshit about weather modeling instead.
- Also unlike the conclusion says, the real filled mass should be able to be back calculated with decent accuracy (assuming isothermal nitrous mass in the tank) using the LOTS fill pressure data and the flight dip tube calculation. Again, I ran out of time here.
- So long as your future rocket a) has a similar number of fasteners/coupler joints/mismatched body tubes, etc. to LOTS, b) has similar paint prep as LOTS, and c) has approximately the same mass/velocity profile/size as LOTS, these results will probably pretty useful. But if the mass/size, surface conditions, or flight profile (i.e. speed regimes and times spent there) change a lot in a future rocket, you might have to go back to square one and use your feelings-based engineering to pick a 'representative' roughness height value as the 1 μm measured roughness is obviously too optimistic.

September 21st, 2023
Aaron Leszkowiat

A Practical Method For Reducing Skin Friction Drag in Sounding Rockets

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Prof William Owen
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Dear Professor Owen,

This report, entitled “A Practical Method For Reducing Skin Friction Drag in Sounding Rockets”, was prepared as my 4A work term report. The report describes a practical way for amateur rocketry organizations such as the Waterloo Rocketry team at the University of Waterloo to make their rocket flights much more efficient without requiring complex analytical techniques or expensive materials. The intent of the report is to lead the reader through the design ideation process that allowed a method to be selected, researched, implemented, and evaluated.

This report was written entirely by me, and has not received any previous academic credit at this or any other institution. I would like to acknowledge the help of Joel Godard for his leadership of the flight dynamics subsystem on the Waterloo Rocketry design team and his support in carrying out the research and decision-making described in the report. I would additionally like to acknowledge Henry Ma and the entire Multi-Scale Additive Manufacturing laboratory for their instrumental assistance with providing the sensitive measurements required to accurately characterize the samples as described in the report. Furthermore, I would like to sincerely thank Teresa Tang for her tireless efforts in assisting me with research and prototyping surrounding the topics of painting, vinyl wrapping, sanding, and polishing. Finally, I would like to express my deepest gratitude to Teresa, Delaney Dymont, and numerous other team members who poured weeks of their free time into making a very special sounding rocket look as smooth and shiny as possible.

Best Regards

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Summary

This report describes a practical method for amateur rocketry organizations to significantly increase the apogee of their rockets. The selected physical mechanism for the apogee increase is a reduction in skin friction drag owing to the smooth surfaces of the rocket. Detailed research was carried out to select the most practical means of achieving this, and it was determined that careful painting and sanding could achieve the desired results with minimal cost and complexity. Furthermore, detailed measurements of painted surfaces were taken using a laser microscope to accurately characterize the true surface properties of the paint and thus quantify the associated effects on the skin friction drag. A detailed methodology is also included with the hope that the results will be replicable for the team in the future, or for other amateur rocketry organizations.

1.0 Introduction

Waterloo Rocketry is a student design team based out of the University of Waterloo in Waterloo, Ontario, Canada. Waterloo Rocketry designs, builds and launches experimental sounding rockets to target altitudes as high as 30,000 ft in intercollegiate rocketry competitions in the United States and Canada. The team gives students an opportunity to develop practical engineering experience by working on numerous student-developed subsystems. Some examples of current subsystems on the rocket and its ground support equipment are engines and fluid systems, composite aerostructures, scientific experiments in biology and radiation physics, launch towers and test stands, and avionics for sensing, actuator control and radio communication. The team's rocket for the 2022-2023 design cycle was named *Leviathan of the Sky* (LOTS). LOTS was successfully launched to an apogee of 9594 m (31476 ft) on June 21st, 2023 at the Spaceport America Cup in New Mexico, achieving a 2nd-place finish in its category.



Figure 1: *Leviathan of the Sky* shortly after engine ignition.

The scope of work on the team comprises not only research, design and construction of the individual subsystems, but also extensive testing and integration to ensure that the systems work as intended and are able to function together as a cohesive whole. Since the team fields a new rocket each year, timelines have to be managed effectively in order to successfully deliver a design that meets the requirements (imposed by both the competition rules as well as the team's own objectives) and is reliable enough to operate in the adverse conditions and limited time window associated with the launch campaign. Therefore, students are able to gain experience with not only the detailed technical work of their subsystems, but also learn to manage projects, budgets, and schedules, and perhaps most importantly, to work effectively as a team [1].

1.1 The Flight Dynamics Subsystem

There are also many subsystems on the team that do not produce physical products *per se* but are still critical to the functioning of the rocket and the team, such as the business, media and outreach, flight dynamics, and software subsystems. This report focuses specifically on the flight dynamics subsystem, which is responsible for managing the design of the rocket at the highest level. This process begins with setting the key parameters of the flight, such as the target altitude (i.e., the apogee) and required stability of the rocket. Once these goals are set, mass and size budgets are assigned to the different subsystems throughout the rocket. These metrics are monitored throughout the design cycle to ensure that the flight parameters (apogee and stability) will not be adversely affected if the anticipated mass, size or location of a subsystem changes significantly.

A key responsibility of the flight dynamics subsystem is evaluating the aerodynamic impacts of external components of the rocket. Salient examples of such components include the nose cone and fins. However, less noticeable aspects of the exterior of the rocket such paint, seams and exposed fasteners can also contribute significantly to drag and thus are also a topic of interest for flight dynamics analysis.

The flight dynamics subsystem is also responsible for assisting other subsystems with design trade-off studies to optimize the overall design of the vehicle. This allows the relative performance of different designs to be evaluated against the design effort, cost,

and technical risk required to implement them. This process is extremely important given the tight timelines of the annual design cycle. Detailed analysis and testing is not always feasible, so the time and effort required to implement a design must be carefully considered against its technical benefits. These studies are primarily accomplished using OpenRocket, which is an open-source flight simulation software used by the team. The technical details and team usage of OpenRocket are discussed in more detail in subsequent sections of the report.

Rocket design is often very iterative in nature, in that detailed design decisions rely on other detailed design decisions which may not yet exist. Therefore, the key mechanism through which the flight dynamics subsystem performs its responsibilities are a series of reports released throughout the design cycle of the rocket. These start with the gathering of existing design data from different subsystems (such as lengths, total masses, mass distributions, shapes of external components, etc.). Using this data, design trade-off studies are performed in OpenRocket, and the results of these studies are made available to the other subsystems in the form of a report. More detailed design decisions can then be made using the results of previous studies. The earlier reports in the design cycle tend to be focused on informing system and subsystem-level design decisions, while the later reports tend to focus on generating information for future documentation and preparation for the final flight trajectory simulation.

This iterative analysis and reporting process allows the team to catch high-level design issues early on, and then later focus on detailed designs that eventually lead to the finished product. The team has found that this approach of successive, simultaneous refinement of all designs (as opposed to detailed design work done in isolation by each subsystem) is a good way to make sure that a rocket design broadly suitable to all subsystems can be found in the short time available.

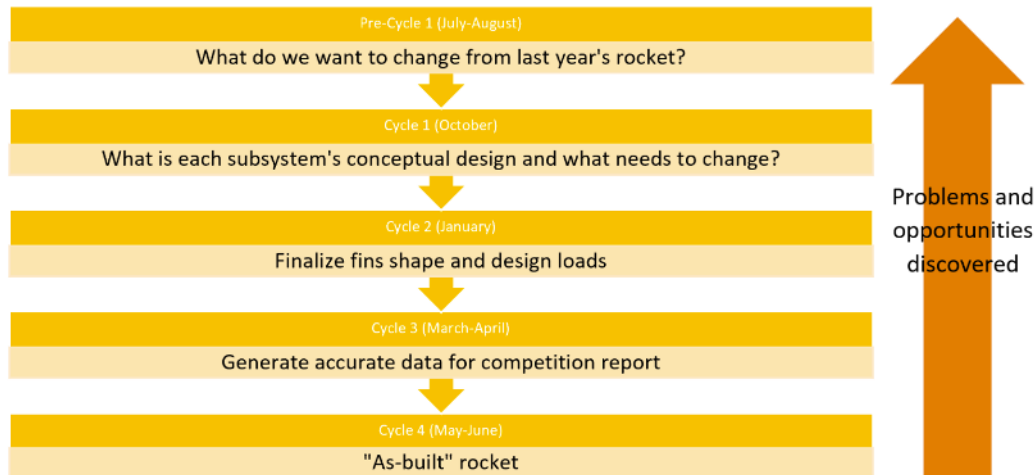


Figure 2: Flight dynamics subsystem workflow [1]

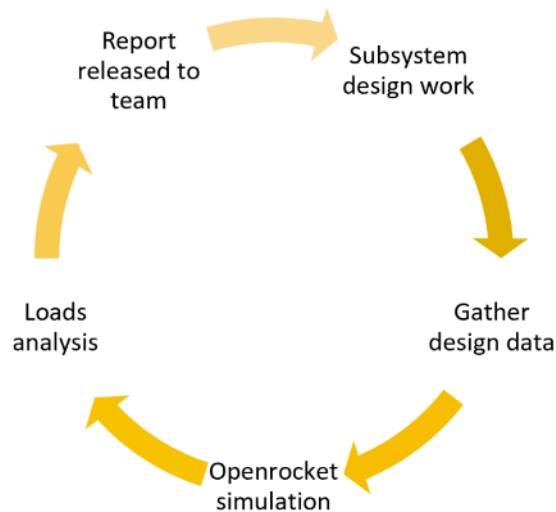


Figure 3: Design and integration cycle workflow [1]

1.2 Design Trade-off Studies as Performed by the Flight Dynamics Subsystem

As mentioned earlier, one of the key responsibilities of the flight dynamics subsystem is to perform design trade-off studies to assist subsystems with their individual designs as well as to optimize the overall design of the rocket. Some design trade-off studies, especially those performed early in the design cycle of the rocket, can be very high-level and far-reaching.

One example of this kind of study from the LOTS design cycle was exploring the effects of changing the diameter of the rocket. Increasing the diameter of the rocket would

make physical packaging of the internal components easier and thus potentially save some design effort later on, but could also increase the drag experienced by the rocket, reducing its apogee. This study was originally motivated by the observation that similar rockets from other universities tended to have larger diameters than those built by Waterloo Rocketry, and the desire to seek a physical explanation for this observation. The design trade-off study found that increasing the diameter significantly impacted the apogee, and therefore the conclusion was made that other teams must have selected this trade-off to use a larger diameter in order to ease their packaging constraints, not for any aerodynamic benefit. Given the team's excellent ability to deal with tight physical packaging, as well as a large existing inventory of materials and equipment compatible with the existing rocket diameter, the decision was made to keep using the same diameter for LOTS as was used for previous rockets.

Another example of a high-level design trade-off was selecting the thrust to be produced by the engine which propels the rocket. A higher thrust might be more difficult to accomplish from the perspective of engine design and impart higher structural loads on the rocket, but such a change would allow the rocket to spend less time in the high-drag transonic flight regime [2], allowing a higher apogee to be achieved with the same amount of propellant. The results of this study indicated that a near-doubling of the peak engine thrust (as compared to the previous value) would yield an optimal trajectory that balanced the flight stability, engine design requirements, structural loads, and aerodynamic drag. This seems like an ambitious design target for the engine, but since the total required impulse to be delivered by the engine was about the same as before, the burn duration was significantly reduced. This change made it easier to accomplish an adequate thermal design of the engine's combustion chamber, and is a good example of how high-level flight dynamics studies affect many different subsystems to optimize the design of the entire rocket as well as its components.

Other design trade-off studies, typically those occurring later in the design cycle of the rocket, take place at lower levels and usually concern implementation details of a particular subsystem. For example, in the LOTS design cycle, some issues were experienced with the fabrication of custom fiberglass body tubes used to construct

sections of the rocket. Since the team was having difficulty with successfully producing a body tube for the upper section of the rocket, a flight dynamics design trade-off study was used to evaluate the impact of using a heavier, commercially-produced body tube instead. It was found that there was no significant negative impact to the rocket's apogee or stability with the commercial body tube, so this option was selected to save time, as other design and fabrication objectives were being blocked by the unavailability of the body tube.

It should be emphasized that the results of design trade-off studies are not universally applicable to all rockets designed by the team in the future. The specific design parameters of the rocket in question, as well as the objectives of a given design cycle, are likely to significantly influence the results of said studies. Therefore, the results of the aforementioned studies should not be taken as 'rules of thumb' and care should be taken to sufficiently analyze the actual system in question for future designs.

1.3 OpenRocket

As aforementioned, the flight dynamics subsystem uses the OpenRocket flight simulation software to analyze and predict the trajectory and performance of the rocket in flight. This open-source software, originally released in 2009 by Sampo Niskanen [3], allows for the computation of various aerodynamic properties of sounding rockets using various well-established models, as well as the simulation of the flight performance and trajectory of the rockets.

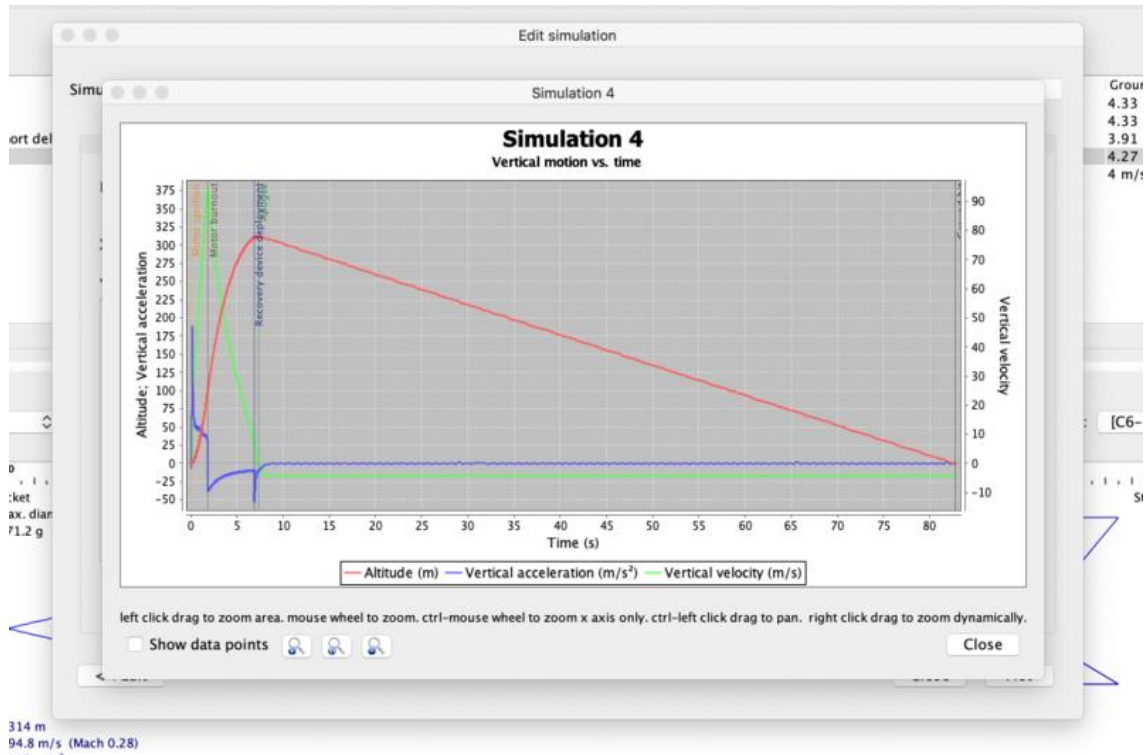


Figure 4: A screenshot of the OpenRocket user interface, showing the trajectory for an example rocket [4]

OpenRocket allows users to benefit from a robust collection of empirical and theoretical models concerning aerodynamics, ballistic trajectories, and other physical phenomena related to the flight of the rocket. Being able to use this kind of detailed modelling without having to implement it from scratch makes OpenRocket incredibly useful, and thus it is the main tool employed by the flight dynamics subsystem. It allows for simplified design trade-off studies to be easily conducted, simply by changing the input parameters to the simulation and observing the effects on the output parameters. It also allows for very detailed simulation later in the design cycle; component masses and engine performance can be tweaked as late as needed in order to precisely hit the target apogee.

In the past, the flight dynamics and software subsystems have collaborated to develop an extension to the OpenRocket source code which allows statistical analysis of the rocket's trajectory. This allows for random, unpredictable variables such as wind speed and direction to be accounted for in a systemic manner when selecting worst-case values for the environmental conditions in which the rocket must operate [1]. In the future, the

flight dynamics and software subsystems will collaborate on further extensions to the OpenRocket source code that will allow for simulation of more complex vehicle designs with active control surfaces.

1.4 Classification of Drag

One important design trade-off study that was proposed early in the LOTS design cycle was an exploration of possible options to reduce the aerodynamic drag experienced by the rocket. This would be valuable since decreasing the aerodynamic drag would allow a higher apogee can be achieved for the same amount of impulse delivered by the engine, easing design restrictions in a number of different subsystems. The drag forces acting on a body moving through a fluid (such as air) can be broadly separated into pressure drag and friction drag. Pressure drag is caused by imbalanced pressure forces acting on an object, typically due to separation of the fluid flow over the object, whereas skin friction drag is related to the viscosity of the fluid in the boundary layer around the object [5]. For a blunt object, pressure drag tends to be the dominant effect, but for a streamlined body, skin friction drag becomes more important [6].

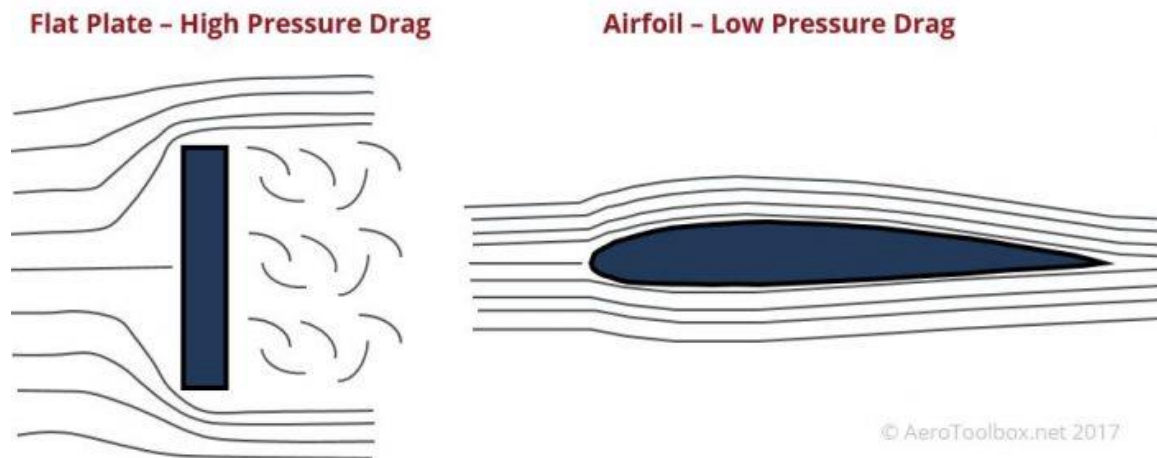


Figure 5: Illustration of the physical mechanism of pressure drag [7]

Thus, the first step in reducing the overall drag acting on the rocket is to give it a more streamlined shape. The design of the components at the front and rear of the rocket (i.e., the nose cone and boat tail) is therefore important in reducing pressure drag by allowing the airflow to closely conform to the rocket. Nose cone design is a very well-studied topic and numerous solutions exist for different criteria and conditions. For LOTS,

the Von Karman nose cone shape was used [1], which minimizes drag for a given length and diameter [8]. However, the design of the boat tail is more constrained than that of the nose cone due to manufacturing and engine packaging constraints, so a simple conical shape was selected for LOTS to save time in design and manufacturing.

1.5 Objective

As detailed above, most of the solutions for reducing pressure drag that could have been reasonably implemented for LOTS were already known from and implemented for previous designs as well. In other words, the LOTS design had a relatively streamlined shape from the initial stages, so focusing on the reduction of pressure drag would probably involve a significant amount of effort for a small amount of apogee gained. Therefore, the prediction and reduction of skin friction drag became an overarching focus for flight dynamics throughout the entire design cycle.

By focusing on reducing the skin friction drag acting on the rocket, the total aerodynamic drag was able to be reduced, allowing the rocket's mass to increase without affecting its apogee. These less restrictive mass requirements for the rocket meant it was easier to design, so minimizing the skin friction drag as much as reasonably possible is recommended for high-performance rockets built by the team in the future. This report describes the research, selection, and implementation for a practical and predictable method which was used to reduce the skin friction drag acting on the rocket.

2.0 Methods for Reducing Skin Friction Drag

There are numerous practical ways to reduce the skin friction drag acting on the rocket; a few are discussed in the following section. In order to select the best one(s), the difficulty of implementation had to be compared with the expected reduction in drag.

2.1 Turbulence Onset

One method to reduce the skin friction drag would be to delay the onset of turbulence in the airflow across the rocket, since the skin friction drag is smaller for laminar flows [9]. Unfortunately, accurately predicting the onset of turbulence is difficult, and the team does not have much experience with the computational fluid dynamics analysis required to predict it or its effects. Furthermore, it is not clear that there exists a reliable means of controlling the turbulent onset, given the sensitivity of the transition phenomenon [10] to the practical constraints of manufacturing tolerances and surface consistency. Therefore, it was decided that attempting to control the onset of turbulence would not be a feasible approach to reduce the skin friction drag.

2.2 Surface Finish

Another method to reduce the skin friction drag would be to simply make the surface of the rocket as smooth as possible. An object with a smooth surface will experience less skin friction drag than those with a rough surface [11]. Therefore, if a simple method of making the rocket significantly smoother was found (e.g., sanding the rocket's exterior surface to a much finer finish), it might represent an easy way to significantly reduce the skin friction drag without requiring sophisticated analysis or precise fabrication.

2.3 Effect of Surface Finish on Apogee

To verify this hypothesis and explore the effects of the surface finish on the skin friction drag, simulations were conducted in OpenRocket. Here, the apogee of the rocket was used as a proxy for the skin friction drag. Although the drag force data can be exported from OpenRocket, it is more useful from a rocket design perspective to discuss the effects in terms of apogee gained or lost. OpenRocket allows a surface roughness to be individually set for each component in the model, but for this exercise, it was assumed

that component roughness is uniform across the rocket. The table below shows the possible types of surface finish which can be selected in OpenRocket:

Table 1: Options for component surface finish in OpenRocket

Surface Finish Option	Roughness Height (μm)
Rough	500
Rough unfinished	250
Unfinished	150
Regular paint	60
Smooth paint	20
Optimum paint	5
Aircraft sheet-metal	2
Finished/polished surface	0.5
Mirror	0

It should be noted that the term ‘roughness height’ has a slightly different meaning than the terms ‘surface roughness’ or ‘surface finish’. The distinction will be important later, but for the purposes of exploring the effects of the surface finish on the apogee of the rocket, the roughness height in units of micrometers (μm) will be used.

It should also be noted that the default value for surface roughness height used by OpenRocket is 60 μm . As far as it is known to the rocketry team’s internal documentation, the default value had always been used for all previous OpenRocket simulations done by the team up until this point.

The input parameters (besides the surface finish) of the OpenRocket simulations to examine the effect of surface finish on apogee are shown in Table 2 below. The mass, size, and shape of the rocket used in the study was the same for every surface finish case examined, and were the same as those of *Kraken of the Sky* (KOTS), the team’s rocket from the previous design cycle. Because the KOTS and LOTS were anticipated to have roughly the same dry mass, size, and shape, this allowed a representative understanding to be gained without detailed design data about the new rocket in hand. Engine performance data (propellant mass and total delivered impulse) was taken from ‘Static

Fire 7', the most recently available test firing of the Kismet v4 hybrid rocket engine intended to power LOTS.

Table 2: Input parameters for an OpenRocket simulation to explore the effect of surface finish on apogee

Input Parameter	Value	Units
Dry mass	34.337	kg
Overall length	425	cm
Overall diameter	15.2	cm
Propellant mass	19.282	Kg
Engine Impulse	36198	N*s
Fin configuration	(Identical to KOTS)	N/A
Nose cone configuration	(Identical to KOTS)	N/A

Using these input parameters, simulations were run in OpenRocket for every possible type of surface finish. The results in terms of apogee are shown in the table below:

Table 3: Effect of surface finish on apogee as predicted by OpenRocket

Surface Finish Option	Apogee (m)
Rough	8292
Rough unfinished	8775
Unfinished	9126
Regular paint	9737
Smooth paint	10430
Optimum paint	10830
Aircraft sheet-metal	10831
Finished/polished surface	10831
Mirror	10831

From the above data, it can be seen that there appears to be diminishing returns on improving the surface finish of the rocket, in that the smoothest surfaces display no

improvement in apogee past a certain point within the significant figures reported by the program.

Importantly, it can also be seen that the surface finish does indeed have a large effect on the skin friction drag and thus the apogee achieved by the rocket. OpenRocket predicted a total difference of 2539 m (8330 ft) between the apogees associated with the roughest and smoothest surface finishes, and a difference of 1094 m (3589 ft) between the default surface finish (60 μm , i.e. 'regular paint') and the smoothest surface finishes.

2.4 Affect of Surface Finish on Stability

As mentioned briefly earlier, one of the key flight parameters of the rocket is its stability. This is essentially a measure the rocket's capacity to correct against unbalancing forces (asymmetric mass distributions, variations in thrust, wind, etc.). While large rockets are actively stabilized by means of computer-controlled thrust vectoring, this is usually impractical for sounding rockets and therefore passive stabilization is used. This type of stabilization relies on the use of fins to shift the rocket's center of pressure below its center of gravity, allowing a restoring torque to be generated when imbalance occurs [12]. Therefore, any design change which shifts the center of gravity towards the front of the rocket, or shifts the center of pressure towards the rear of the rocket, or shifts both centers simultaneously, will have a positive effect on the stability. However, if the stability is too large, the rocket is considered over-stable as it will tend to align itself with the wind, reducing the maximum apogee achieved by the rocket [13].

Another key finding of the OpenRocket simulations which were described in the previous section was that the surface finish had no effect on the stability of the rocket. This meant that any project to improve the surface finish of the rocket could be undertaken more or less in isolation, without needing to consider its effect on stability. Since corrections to the stability can sometimes require relatively significant design changes (such as adding or subtracting mass from components, or changing the shape and size of the fins), it was useful to know that there were no dependencies between the final surface finish that might be achieved and the design of the rest of the rocket. This

independence is physically explainable by the fact that, by assuming a uniform surface finish across the entire rocket, any change to the surface finish would affect the skin friction drag on all areas of the rocket equally and thus have no effect on the center of pressure.

2.5 Significance of Surface Finish Apogee Effects

To develop a better sense of how significant the aforementioned improvements to the rocket's apogee were, more OpenRocket simulations were performed. The surface roughness height was kept the same at 60 μm , but the dry mass of the rocket was varied in 1 kg increments in both directions. Since the location where the mass is added or subtracted does have an affect on the stability and thus apogee as previously described, the mass should be added or subtracted as close as possible to the existing center of mass. Conveniently, the component representing the rocket's oxidizer tank in the existing OpenRocket model of KOTS was positioned approximately centrally about the existing center of mass. Therefore, the mass of the oxidizer tank was modified to examine the effect of changing the rocket's mass without significantly affecting its stability, allowing a more representative comparison to be made. The results of this exercise are shown in Table 4 below:

Table 4: Effect of mass on apogee as predicted by OpenRocket

Dry mass (kg)	Apogee (m)
33.337	9846
34.337	9737
35.337	9624

For context, achieving a mass reduction of 1 kg using the design of the previous rocket (KOTS) as a baseline would certainly be feasible, but saving larger amounts of mass would become increasingly difficult. Large mass savings would require more effort in both design and analysis, as well as more expensive materials such as high-strength 7000-series aluminum alloys, and/or more sophisticated manufacturing techniques such

as CNC machining. Such efforts would obviously become increasingly costly in terms of both time and money.

It can be seen from the above data that decreasing the dry mass of the rocket by 1 kg is predicted to increase the apogee by approximately 109 m. As a verification of the simulation, increasing the mass by the same amount decreases the apogee by roughly the same amount. The relationship between rocket mass and apogee is not expected to be linear when extended to larger values of mass added or subtracted, but given the practical upper limits of mass reduction as previously described, varying the mass over this relatively small range still serves as a useful benchmark.

To summarize and contextualize the results of these OpenRocket design trade-off studies, it was found that that reducing the dry mass of the rocket by 1 kg is expected to increase its apogee by 109 m. In comparison, the apogee increase expected to be achievable by sufficiently refining the surface finish of the rocket is more than an order of magnitude larger, at 1094 m. Thus, to achieve a similar improvement to apogee, the new rocket design would require a dry mass reduction on the order of 10 kg, which would likely be very difficult to accomplish. Therefore, it is clear that improving the surface finish, even modestly, could make a large increase in apogee relatively simple to achieve.

2.6 Types of Surface Improvements

Based on the design trade-off study described in the previous sections, it was concluded that improving the surface finish of the rocket was a promising option to minimize the skin friction drag acting on the rocket. Therefore, at this point in the design cycle of the rocket, the flight dynamics subsystem shifted its focus from predicting the effects of changing the surface finish to exploring practical methods for actually achieving the required improvement.

The first option considered was to simply sand the exterior surfaces of the rocket to a very fine finish. All of the large exterior surfaces of the rocket are either painted composite materials (the fin can, combustion chamber body tube, oxidizer tank aft skirt, upper body tube, parachute tube, and nose cone) or bare aluminum (the boat tail and oxidizer tank). Both of these surface types are suitable to fine sanding, making this a

feasible option that could be accomplished without much cost or complexity. However, this option would be relatively labour-intensive due to the large amount of sanding expected to be required to achieve a sufficiently fine finish, so time would need to be set aside in the fabrication schedule in the rocket for this purpose.

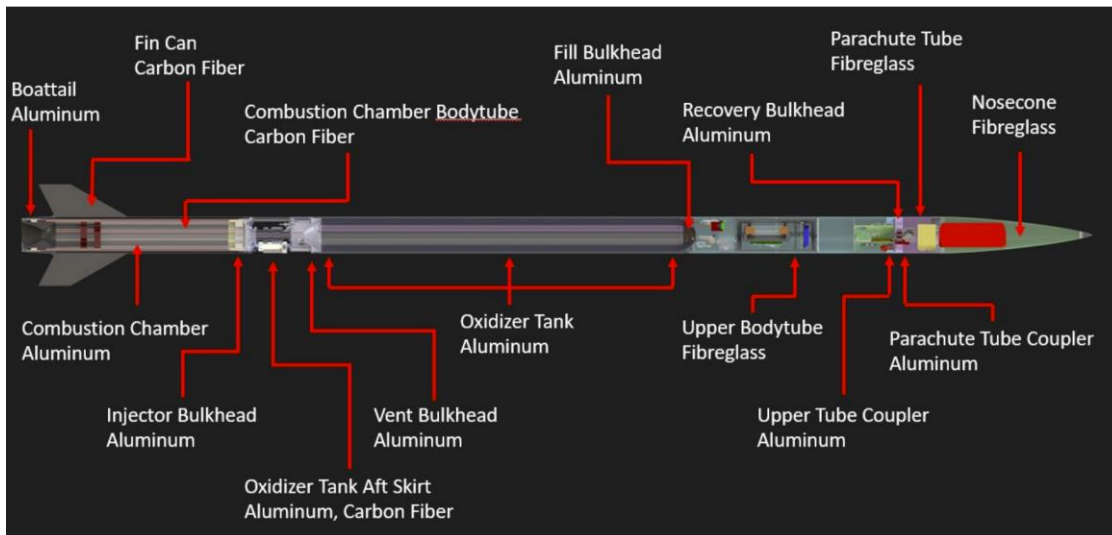


Figure 6: Physical layout of LOTS showing the location and size of exterior components [1]

The next option considered was the use of a commercial epoxy resin system to coat the exterior surfaces of the rocket. Several products specifically designed to produce smooth, durable coatings exist and are used by amateur rocketeers for this purpose [14]. This approach might be a relatively low-effort way to produce a smooth surface, but several downsides exist. For one, an epoxy coating would be more expensive than the exterior paint already used by the team. Another is that if the team were to rely on a specialty coating, it could become challenging to source an alternative if the selected product became unavailable; another specialized material added to the ‘supply chain’ of the rocket represents additional complexity and risk. Additionally, it was expected that an epoxy coating would be thicker than an equivalent coating of paint, thus adding additional mass to the rocket when compared to the paint option.

The final option considered was the use of a vinyl wrap. This was expected to have a very smooth surface finish, and like the epoxy coating, would probably require less manual labour by team members to achieve a good result. Another benefit of using a vinyl wrap is that the aesthetic designs and sponsor logos which need to be added to the rocket

can simply be printed onto the vinyl, avoiding the need for multiple coats of coloured paint or epoxy that would be required to achieve the same level of detail. However, like the epoxy coating, it also represented a sourcing challenge to the team, in that it would be more expensive to implement than paint and could pose issues with availability and/or lead time. Furthermore, while paint or epoxy would be relatively easy to repair if scratches or damage were to occur, there is no obvious method of repairing the vinyl wrap surface if it were to become damaged. One additional issue that was anticipated with the vinyl wrap approach is that application onto surfaces that do not have regular curvature, such as the fins or nose cone, would be difficult or even completely infeasible.

For the reasons described above, it was decided that simply sanding the existing exterior surfaces of the rocket would be the simplest, cheapest and overall most suitable method to achieve the desired improvements to the surface finish of the rocket.

2.7 Lack of Roughness Height Data

Having determined the most suitable method for improving the surface finish of the rocket based on the logistical and technical constraints of the rocket design, work was started on the implementation details. However, one problem quickly arose: there was no straightforward way to estimate the surface roughness height. Without knowing the surface roughness height of the paint, there would be a very large amount of uncertainty in the performance of the rocket. As shown in the previous OpenRocket simulations, the difference in apogee between rough and smooth paint can easily be many hundreds of meters. Since the competitions in which the team participates judge performance based on how close the rocket's actual apogee conformed to the target or predicted apogee, it is necessary to not only minimize the skin friction drag as much as possible, but also to accurately characterize it so that reliable flight simulations are possible.

An extensive search for relevant literature on the surface finish properties of painted surfaces was carried out, but unfortunately very little useful data was found in the public literature describing the manufacturing processes for paints and not simply the finished-result surface roughness height. Nothing was found which made it possible to estimate the surface roughness height for the combination of sanding processes and

paints that the team was considering using, and there was little else for any kind of sanding or painting process at all. The specific painting and sanding processes that were settled upon will be described in detail in a subsequent section, but since the literature review did not find any useful data, it was deemed necessary to prepare representative samples of the paint desired to be used on the rocket and take measurements to acquire the necessary surface roughness data.

3.0 Surface Roughness Measurement

Before samples could be prepared and measured, it was necessary to determine which kinds of paint and sanding processes were likely to produce a sufficiently smooth surface while minimizing the cost and effort required.

3.1 Selection of Painting Materials and Techniques

In the past, the team has used TREMCLAD® oil-based spray paint for the painted surfaces of the rocket. While this type of paint was convenient to apply, readily available and bonds well to the composite materials being painted, it is difficult to apply consistently. Furthermore, its color cannot easily be changed by mixing and its viscosity or thickness cannot be modified through the use of solvents. For this reason, it was desired to use a more traditional paint, which can be modified as needed and applied more consistently.

After some research about painting techniques, a method was found that produced a smooth, consistent surface with a minimal amount of work. The technique is to use a solvent to dilute the paint before application. By applying the diluted paint with a sponge brush in thin coats and coarsely sanding in between coats to promote surface adhesion, a smooth, even surface is produced that can then be easily sanded to the required finish [15]. The paint chosen was TREMCLAD® oil-based rust paint, since it is inexpensive, readily available in the desired colours, and is compatible with the materials being painted. The 'semi-gloss' type was selected to balance visual appearance with susceptibility to marking or damage. This painting technique was evaluated by the team members responsible for painting by the rocket and was found to be feasible, so work began on preparing samples for measurement.

3.2 Sample Selection and Preparation

Eight different samples comprising four unique types were prepared in accordance with the paint research and experimentation described above. The list of samples is given in Table 5 below:

Table 5: List of surface samples prepared for measurement

Sample ID	Description	Size
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A1	White on black paint	Large
A2	White on black paint	Small
B1	Black paint	Large
B2	Black paint	Small
C1	Vinyl	Large
C2	Vinyl	Small
D1	Bare aluminum	Large
D2	Bare aluminum	Small

Two different sample sizes of each surface type were produced in order to ensure that the samples would be physically compatible with the measurement device selected. Two different types of painted samples were prepared (A and B) to see if the colour or application order had an effect on the surface roughness. Additionally, vinyl samples were prepared to take advantage of the measurement opportunity, even though it was decided that vinyl would not be used on the rocket. Finally, samples of bare aluminum were prepared to represent the unpainted aluminum surfaces of the oxidizer tank and boat tail.

The painted samples were prepared on a fiberglass substrate. The paint was diluted at a 9:1 volumetric ratio of paint to mineral spirits and was applied using sponge brushes. In order to promote paint adhesion, the substrate was first sanded to 600 grit and then cleaned with a detergent. Next, coats were applied until the desired visual appearance was achieved; each coat was allowed to cure and then sanded to 600 grit and cleaned with detergent before applying the next coat. All painted samples had between 4 and 6 total coats. The final coat was sanded to 2000 grit, polished using Meguiar's® Ultimate Compound applied with a soft cotton cloth, and finally waxed using Turtle Wax Super Hard Shell Car Wax Paste.

The vinyl samples were placed on a clean aluminum substrate, but no specific preparation materials or techniques were otherwise used. The aluminum samples were sanded to 2000 grit and finished with polish and wax in a similar manner to the painted samples.

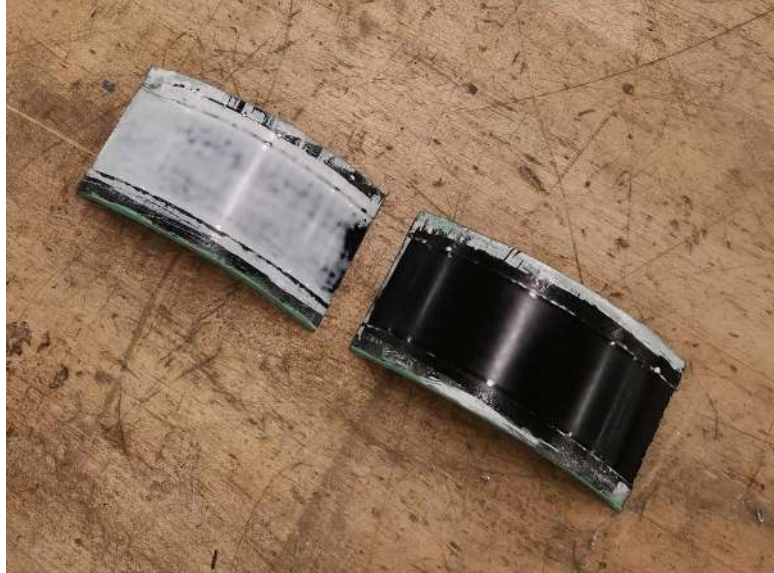


Figure 7: Prepared samples A1 and B1

3.3 Measurement

There are many different methods of measuring the roughness of a surface, so a search was carried out for potential partners who would be willing to work with the rocketry team to take precise measurements of the prepared samples that were required. Eventually, a suitable partner was found in the Multi-Scale Additive Manufacturing (MSAM) laboratory, which is hosted by the University of Waterloo. MSAM was able to characterize the samples using a VK-X210 laser microscope, which is capable of measurements of up to 0.5 nm resolution in roughness height [16]. After the samples were sent to MSAM, the measurements were performed without issue and the data was received. Surface profile data (height versus positional index) were obtained at three unique points each on four of the samples sent (A1, B2, C1, D2) for a total of 12 different sample points. The figure below shows a representative example of the visual appearance of the data when plotted.

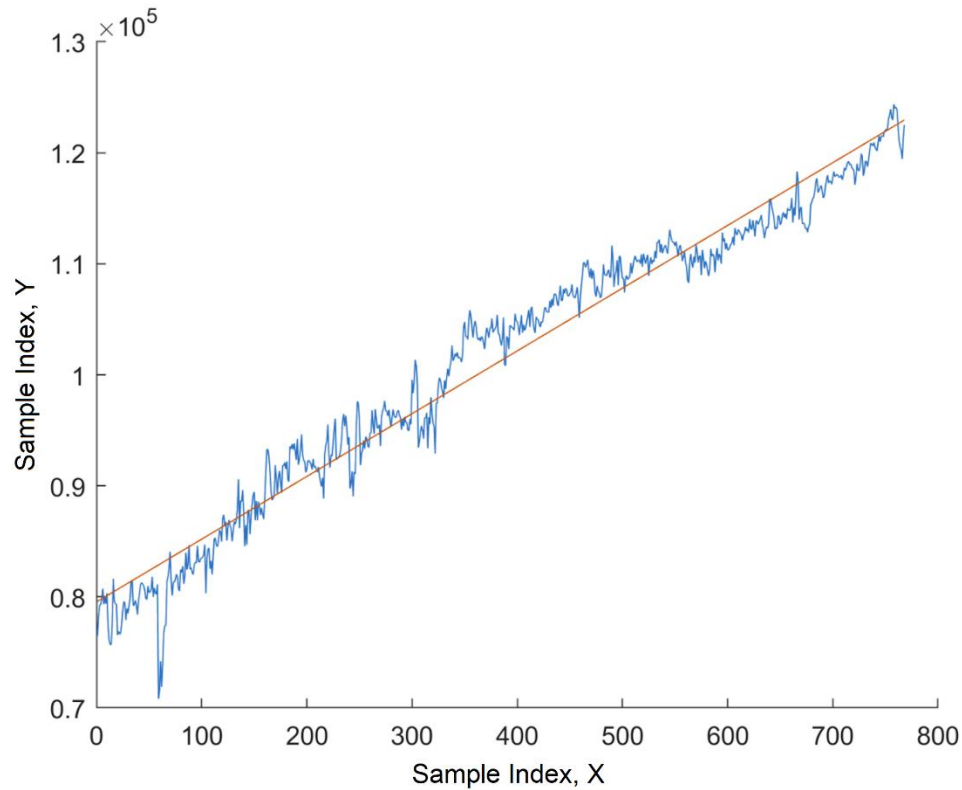


Figure 8: Example surface profile data with linear profile fit

3.4 Units and Methods for Evaluating Surface Finish

As mentioned earlier, there is a distinction between the terms ‘surface roughness’ and ‘roughness height’. While the two terms do ultimately refer to the same property of a surface (stated simply, how ‘smooth’ it is), they are used in different contexts and conversion factors must be used to translate between them.

The terms ‘surface roughness’ or ‘surface finish’ are typically used in a manufacturing context, with the goal of controlling a surface’s smoothness usually for sealing, locating, or aesthetic purposes. The values calculated by the various methods differ in aspects such as their sensitivity to extreme values, so different calculation methods are used for different applications. Two different calculation methods for surface roughness are described below. Each method is theoretically defined assuming a continuous height function of the surface, but in practice discrete measurements are used

for these calculations and therefore digital approximations of the calculation methods are used instead [17].

Calculating Ra: $Ra = (1/L) \int_0^L Z(x) dx$ L = evaluation length Z(x) = the profile height function The digital approximation is: $Ra = (Z_1 + Z_2 + \dots + Z_N) / N$	Calculating RMS: $Rq = \left[(1/L) \int_0^L Z(x)^2 dx \right]^{1/2}$ L = evaluation length Z(x) = the profile height function The digital approximation is: $Rq = \left[(Z_1^2 + Z_2^2 + \dots + Z_N^2) / N \right]^{1/2}$
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Figure 9: Continuous and discrete calculation methods for Ra and RMS surface roughness

The profile height function Z(x) is essentially a ‘mean line’ of the surface profile, against which the surface deviations can be measured. However, generating this function is not necessarily simple, as the theoretically ‘ideal’ surface profile is not necessarily a simple shape such as a plane. Additionally, there usually exists a larger-scale pattern to the surface profile called ‘waviness’, which must be accounted for to produce useful roughness measurements [18]. This is illustrated in Figure 10 below:

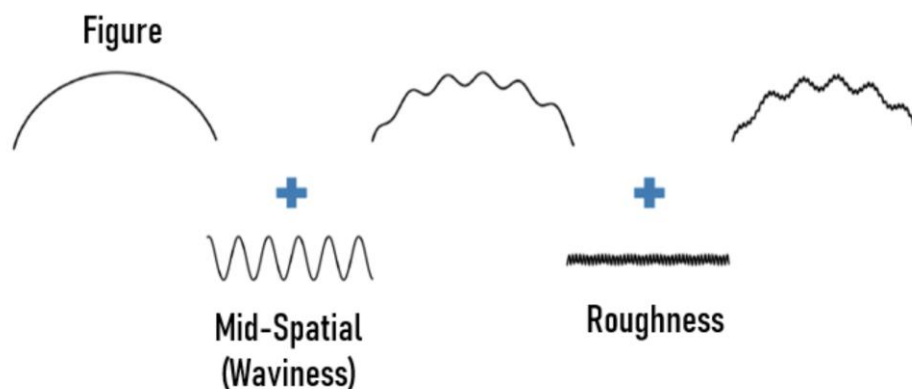


Figure 10: Geometric components of surface texture [17]

In engineering practice, industrial standards such as ASME B46.1 or ISO 4287-1 define the exact methods for determining the profile height function and therefore accurately calculating the surface roughness.

On the other hand, the term ‘roughness height’ is used almost exclusively in fluid mechanics to describe the roughness of a surface in relation to its effect on phenomena such as fluid friction coefficients, boundary layers, and so forth. Reference [19] describes the concept of roughness height (referred to therein as ‘sand-grain roughness’):

The experimentalists who generated the data correlated in the Moody Chart, however, roughened pipe surfaces by coating their internal surfaces with a monolayer of sand, the pipe wall roughness being defined as the average diameter of the sand grains. Thus, the sand-grain roughness values required for use with the Moody Chart are not derived from any direct measure of roughness using modern surface characterization equipment, such as an optical profilometer. Using direct measurements of surface roughness in fluid flow calculations may therefore result in significant error.

Since OpenRocket draws its aerodynamic models from fluid mechanics literature which exclusively uses the roughness height, it is necessary to convert the calculated surface roughness as derived from the experimental paint sample measurements into a roughness height, so that the most representative option from OpenRocket can be selected. Thankfully, reference [19] gives an simple approximate conversion factor between several common types of surface roughness measurements and the surface roughness height. Surface roughness calculated as an Ra value is described as being the most accurate in terms of conversion to roughness height, so this calculation method and its associated conversion factor was used. Therefore, the roughness height ϵ in μm can be calculated from the surface roughness Ra in μm according to the relationship $\epsilon = 5.863Ra$.

With the sample data acquired and a method of converting the calculated surface roughness into a roughness height established, post-processing of the sample data was performed.

3.5 Data Post-Processing

Since the samples were measured at such a high resolution, it was convenient to use a MATLAB script to handle the large number of calculations required to post-process the data. Each measurement point comprised a grid of 1024 recorded height values along the X-axis and 768 values along the Y-axis. As stated previously, three point grids were generated at different unique locations on each of the four samples measured.

For simplicity, it was decided to use a polynomial fit to establish the surface profile height function. Polynomial fits of several different degrees were evaluated, and it was found that a degree-1 polynomial (straight line) fit produced the largest values for surface roughness, so this method was chosen for conservatism.

To calculate the overall average surface roughness for each sample, the surface roughness was calculated using the *Ra* method, working along the X-axis of the measurement grid. This was done for each index of the Y-axis for a given measurement point, generating 768 calculated surface roughness values. The arithmetic mean of these values was then taken to obtain the average surface roughness for each measurement point. No difference was observed when calculating in the reverse manner (calculating roughness values along the Y-axis and then averaging along the X-axis indices). This process was then repeated for the other two measurement points on the same sample, allowing the arithmetic mean of the three measurement points to be calculated. This final mean surface roughness was then converted into a roughness height using the methods described previously.

3.6 Measurement Results

The results of the roughness height calculations as described in the previous section are shown below for each of the four samples:

Table 6: Final calculated roughness heights for each sample

Sample	Roughness height (μm)
A1 (White on black paint)	0.8624
B2 (Black paint)	0.9545

C1 (Vinyl)	0.6426
D2 (Bare aluminum)	1.2518

As shown by this data, all of the prepared samples were very smooth, falling between the OpenRocket options of 'Aircraft sheet-metal' ($2\text{ }\mu\text{m}$) and 'Finished/polished surface' ($0.5\text{ }\mu\text{m}$).

4.0 Flight Preparation and Simulation

With the results of the roughness height measurements in hand, the next task of the flight dynamics subsystem was to implement the painting procedures that had been developed and prepare a final OpenRocket simulation for the rocket's actual flight.

4.1 As-Implemented Surface Finish

Painting the rocket was a labour-intensive activity as anticipated. Thankfully, through the use of careful scheduling, painting was able to occur around the other testing and integration activities requiring the painted components, so no major delays or issues occurred. A representative component of LOTS (the upper body tube section), showing the surface finish after final painting, sanding, polishing and waxing, is shown in Figure 11 below:

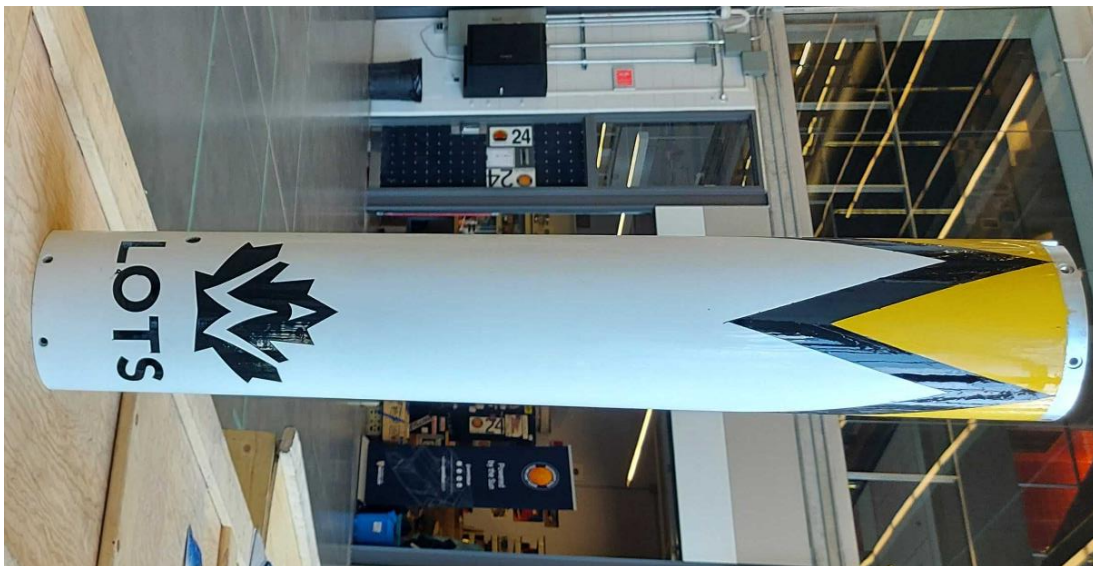


Figure 11: LOTS body tube after final surface processing

4.1 Final Flight Simulation

For the purpose of accurately predicting the apogee of the rocket, it was important to apply some engineering judgement to the calculated roughness height results. First of all, the measurement samples were prepared under ideal conditions and were small in size; it is unlikely that the paint over the entire surface of the rocket was as smooth as the samples, especially in harder-to-reach areas or on components where the painting process was rushed to speed up the fabrication schedule. Additionally, there are numerous external imperfections (mismatched diameters, seams between components, etc.) and other external features (fasteners, umbilical ports, etc.) that would cause the overall drag behaviour of the rocket to deviate from the value calculated by OpenRocket. Since it is not feasible to capture all of these apogee-reducing effects discretely in OpenRocket, the most straightforward option was to simply use a larger value for the roughness height, instead of using the closest option of 0.5 μm .

The selected value was selected semi-arbitrarily using a healthy dose of engineering judgement. There were also non-technical factors that played into the selected value, such as the team's desire to overshoot the target apogee rather than undershoot it, and the likeliest launch conditions based on the constraints of the competition schedule and launch windows. Taking all of this into account, a roughness height of 5 μm was chosen to be used in the final flight simulation.

The final variable that can be tuned after the rocket is designed and built is the amount of oxidizer that is loaded onto the rocket before flight. More oxidizer will cause the engine to produce more impulse, so the predicted apogee can be tweaked until the last practical moment in this manner. Taking the most detailed and up-to-date OpenRocket model of LOTS that existed at this point in the design cycle, as well as the chosen roughness height of 5 μm , allowed the final flight simulation to predict an apogee of exactly 9144 m (30000 ft) with an oxidizer fill mass of 15.63 kg.

5.0 Flight Results and Conclusion

Leviathan of the Sky was successfully launched on June 21st, 2023, achieving an apogee of 9594 m (31476 ft). This apogee is somewhat higher than the predicted apogee of 9144 m (30000 ft), but was still an excellent result, achieving the second-closest result to the target apogee in its category in the history of the Spaceport America Cup.

5.1 Discussion of Apogee Overshoot

Two main explanations are proposed for the observed overshoot in apogee. The first explanation is that the amount of oxidizer filled was more than expected. The maximum amount of oxidizer fill is controlled by adjusting the length of a dip tube inside the oxidizer tank. Provisions were made early in the LOTS design cycle for a load cell integrated within the launch tower that would allow for an accurate measurement of the loaded oxidizer mass, but unfortunately this system did not function as intended and no useable mass data was recorded. Therefore, for the purposes of this analysis it is assumed that the actual mass of oxidizer filled was the theoretical value of 15.63 kg as set by the length of the dip tube.

The second proposed explanation for the overshoot in apogee is that the effective surface roughness height of the rocket (including fasteners, seams, etc.) was in fact lower than the semi-arbitrarily selected value of 5 μm . However, as shown in Table X, the predicted change in apogee between the selected value of 5 μm and the next-lowest value of 2 μm is negligible, so other physical mechanisms (e.g. variation in the actual weather conditions as opposed to those used in the OpenRocket simulation) are probably required to explain the deviation.

5.2 Conclusions

In conclusion, the method proposed to reduce the skin friction drag acting on the rocket appears to have worked well, and also adequately met the original requirements for design simplicity and implementation cost. Additionally, the apogee of the rocket was able to be predicted to within 1.05% of the actual value, which is remarkable considering the large amount of possible variation in the input conditions to the simulation. Overall,

the selected method for reducing the skin friction drag was found to be practical and is likely to be used for future rockets built by the team which require high performance under tight technical and nontechnical constraints.

5.3 Recommendations

It is recommended that more work be done to come up with a more thorough explanation for the observed deviation in apogee. For example, research into the methods used by OpenRocket to model high-level atmospheric winds and other atmospheric effects could suggest an extension to the program which would allow these phenomena to be accounted for more accurately. Additionally, it is recommended that further engineering effort be put into a means to accurately measure the filled oxidizer mass onboard the rocket in order to remove this source of uncertainty from future simulations and flights.

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